

DOCUMENT 14

Troubleshooting Guide

Diagnose and resolve issues with CLI commands, SQL queries, and log analysis

Common Issues — Quick Reference

Symptom	Likely Cause	First Action
Recommendation confidence < 40%	Image quality / heavy grading	Use clean unedited reference
Wrong pattern detected	Mixed lighting or color grade	See Image Quality section below
Shoot Mode ceiling warning	Ceiling height not in Settings	Settings set ceiling height
Kit gear not in suggestions	Gear missing from My Kit	My Kit add missing item
Lab returns 403	Email not in allowlist	Add email to LAB_ALLOWED_EMAILS
Analysis > 30 s	VLM timeout	Check API key; increase timeout
/health returns 500	DB not initialized	<code>python3 -c "from db.database import init_db; init_db()"</code>
Gold set scores dropped	Recent candidate applied	Check candidates table; revert
Seeded data in live metrics	Signal flags not set correctly	Run hygiene audit SQL query
VLM returns garbled JSON	Model output parsing error	Check LOG_LEVEL=DEBUG for raw output
CORS error in browser	Origin not in ALLOWED_ORIGINS	Add origin to env var; restart

API Diagnostic Sequence

Run these checks in order. Each step narrows the failure surface.

bash — step-by-step API diagnostics

```
# 1. Is the backend alive?
$ curl -s http://localhost:8000/health | python3 -m json.tool
# Expected: {"status":"ok","version":"1.0"}

# 2. Is the database accessible?
$ curl -s http://localhost:8000/health/db | python3 -m json.tool
# Expected: {"tables":5,"schema_version":"...", "ok":true}

# 3. Can you authenticate?
$ curl -s -H "Authorization: Bearer $AI_GATEWAY_API_KEY" \
    http://localhost:8000/api/master-modes
# Expected: JSON list of master modes

# 4. Is Lab accessible?
$ curl -s -H "Authorization: Bearer $AI_GATEWAY_API_KEY" \
    -H "x-lab-email: you@org.com" \
    http://localhost:8000/api/lab/signals/summary
# Expected: JSON with live/seeded/internal counts

# 5. Can VLM be reached? (requires a test image)
$ curl -s -X POST \
    -H "Authorization: Bearer $AI_GATEWAY_API_KEY" \
    -H "x-lab-email: you@org.com" \
    -d '{"image_path":"static/uploads/test.jpg","include_debug_overlay":true}' \
    http://localhost:8000/api/lab/analyze | python3 -m json.tool
```

Log Analysis

The backend logs to stdout in structured format. Key prefixes to filter for:

bash — log patterns and grep commands

```
# Enable verbose logging
$ LOG_LEVEL=DEBUG uvicorn main:app --reload

# All errors in current run
$ grep -E "ERROR|Exception|Traceback" /tmp/ngw-dev.log

# VLM call latencies (ms)
$ grep "vlm_pass_ms" /tmp/ngw-dev.log | tail -20

# Failed Lab access attempts
$ grep "lab_access_denied" /tmp/ngw-dev.log

# VLM timeout events
$ grep "VLM_TIMEOUT" /tmp/ngw-dev.log | wc -l

# Signals inserted today
$ grep "signal_inserted" /tmp/ngw-dev.log \
  | grep "$(date +%Y-%m-%d)" | wc -l
```

Database Diagnostic Queries

sql — database integrity and health checks

```
-- Table row counts
SELECT name,
  (SELECT COUNT(*) FROM sqlite_master WHERE type='table' AND name=m.name)
  AS exists_flag
FROM (VALUES
  ('session_signals'),('gold_sets'),('candidates'),
  ('reference_library'),('user_kits')
) AS m(name);

-- Signal count by source
SELECT signal_source, COUNT(*) AS n
FROM session_signals GROUP BY signal_source;

-- Open candidate queue
SELECT status, COUNT(*) FROM candidates GROUP BY status;

-- Gold set score summary
SELECT status, COUNT(*) FROM gold_sets GROUP BY status;

-- Most recent 5 signals
SELECT id, pattern, outcome, signal_source, created_at
FROM session_signals ORDER BY created_at DESC LIMIT 5;
```

sql — find flag_mismatch rows (seeded marked as live)

```
SELECT id, session_id, signal_source, created_at,
  include_in_metrics, include_in_learning
FROM session_signals
WHERE signal_source IN ('seeded', 'internal')
  AND (include_in_metrics = TRUE
    OR include_in_learning = TRUE
    OR include_in_conversion = TRUE
    OR include_in_cohorts = TRUE);
```

Image Quality Diagnostics

Condition	Confidence Impact	Detection	Fix
B&W; / monochrome	-25%	Chroma saturation < 0.05	Use color original
Heavy color grade	-20%	Hue shift or LUT signature	Export ungraded
No face detected	-30%	Face region score = 0	Ensure face in frame
Resolution < 512 px	-30%	Image width check at ingest	Use ≥ 1024 px wide
Extreme contrast	-15%	Histogram clipping > 5%	Recover highlights/shadows
Multiple subjects	-10%	Multi-face region flag	Crop to primary subject
Environmental clutter	-10%	Background complexity score	Plain background preferred

VLM Troubleshooting

bash — test VLM connectivity independently

```
# Test gateway reachability
$ curl -s -o /dev/null -w "%{http_code}" \
  -H "Authorization: Bearer $AI_GATEWAY_API_KEY" \
  https://api.vercel.ai/v1/models
# Expected: 200

# Test with a simple text prompt (no image)
$ curl -s -X POST https://api.vercel.ai/v1/chat/completions \
  -H "Authorization: Bearer $AI_GATEWAY_API_KEY" \
  -H "Content-Type: application/json" \
  -d '{"model": "anthropic/claude-haiku-4", "messages": [{"role": "user", "content": ...}]}'
```

VLM TIMEOUT CHECKLIST

- VLM_TIMEOUT_SECONDS is set (default 30). Increase to 60 for slow inference.
- VLM_PASS_COUNT is set (default 3). Reduce to 1 for faster debug iteration.
- Check AI Gateway quota in Vercel Dashboard AI Gateway Usage.
- Try a lighter model: set AI_MODEL_STRING=anthropic/claude-haiku-4 temporarily.

Error Codes — Complete Reference

Code	Message	Likely Cause	Action
400	Invalid image format	Unsupported file type	Use JPEG/PNG/HEIC $\leq$ 10 MB
400	Missing required field: pattern	Bad request body	Check API docs at /docs
401	Unauthorized	Missing or expired Bearer token	Refresh token; re-run vercel env pull
403	Lab access denied	Email not in allowlist	Add to LAB_ALLOWED_EMAILS; restart
413	Request entity too large	Image > MAX_UPLOAD_MB	Resize image; or raise limit
422	Unprocessable entity	Schema validation failed	Check field types in body
429	Too many requests	Rate limit hit	Back off; check gateway quota
500	Internal server error	Unhandled exception	Check logs; curl /health/db
503	Service unavailable	Backend not running	Start uvicorn; check process
504	Gateway timeout	VLM call timed out	Increase VLM_TIMEOUT_SECONDS

Common Fix Procedures

Reset a stuck analysis session

sql — clear stuck in-flight sessions

```
UPDATE session_signals  
SET outcome = 'unknown'  
WHERE outcome IS NULL AND created_at < DATE('now', '-1 hour');
```

Rebuild SQLite indexes after large import

sql — rebuild indexes

```
REINDEX session_signals;  
REINDEX gold_sets;  
VACUUM; -- reclaim space after bulk deletes
```

GETTING HELP

- Include: session_id (from URL after analysis), error code, LOG_LEVEL=DEBUG output snippet, and signal_source if the issue is signal-related.
- Open a Lab candidate with type="trust_safety" to track systemic issues for curator review.